

Task 2 – CI/CD Pipeline with GitHub Actions

Objective

The objective of this task was to create an automated CI/CD pipeline for a containerized application using GitHub Actions. The pipeline should automatically build, test, validate Docker images, and publish Docker images to Docker Hub whenever code changes are pushed to the repository.

Project Overview

For this task, I used my existing Notes API project built with Go and PostgreSQL.

The Notes API provides:

- User Authentication using JWT
- Notes CRUD Operations
- PostgreSQL Integration
- Docker Containerization
- Docker Compose Deployment

The project was previously containerized as part of Task 1 and served as the foundation for implementing CI/CD automation.

Technologies Used

- GitHub Actions
 - Docker
 - Docker Hub
 - Go 1.25
 - PostgreSQL
 - Git & GitHub
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Task Requirements

The following requirements were implemented:

- GitHub Actions workflow configuration
 - Automated application build process
 - Automated test execution
 - Docker image build validation
 - Docker Hub integration
 - Automatic Docker image publishing
 - Workflow status visibility
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Workflow Architecture

Pipeline Flow:

Developer Push
↓
GitHub Actions
↓
Build Application
↓
Run Tests
↓
Build Docker Image
↓
Publish Docker Image
↓
Docker Hub

GitHub Actions Workflow

Workflow File:

`.github/workflows/ci.yml`

The workflow executes automatically on:

- Push to main
 - Push to feature branches
 - Pull requests targeting main
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Build and Test Job

Responsibilities:

- Checkout repository
- Install Go
- Download dependencies
- Build application
- Execute tests

Commands executed:

go mod download

go build ./...

go test ./...

Docker Build Job

Responsibilities:

- Validate Dockerfile
- Build application image
- Ensure container image can be created successfully

Command executed:

docker build -t notes-api .

Docker Publish Job

Responsibilities:

- Authenticate with Docker Hub
- Build production image
- Push image automatically

Docker Hub Repository:

aayushshantam/notes-api

Published Tag:

latest

GitHub Secrets

The following repository secrets were configured:

Secret	Purpose
DOCKERHUB_USERNAME	Docker Hub username
DOCKERHUB_TOKEN	Docker Hub access token

These secrets allow GitHub Actions to securely authenticate with Docker Hub.

Validation

The following validations were performed:

- Workflow executes successfully on push
 - Application builds successfully
 - Tests complete successfully
 - Docker image builds successfully
 - Docker image publishes successfully
 - Docker Hub repository receives updated image
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Screenshots

Screenshot 1 – Successful GitHub Actions Workflow

docs/screenshots/github-actions-success.png

Description:

Successful execution of all workflow jobs including build, test, Docker image build, and Docker image publication.

Screenshot 2 – Docker Hub Published Image

docs/screenshots/dockerhub-latest-tag.png

Description:

Docker Hub repository showing the latest image tag published automatically by GitHub Actions.

Repository Information

GitHub Repository:

<https://github.com/shantam-sharma/Notes-API>

Workflow File:

.github/workflows/ci.yml

Docker Hub Repository:

<https://hub.docker.com/r/aayushshantam/notes-api>

Branch Used:

feature/github-actions

Deployed Artifact

Docker Image:

aayushshantam/notes-api:latest

The image is automatically updated whenever the workflow successfully completes.

Conclusion

A complete CI/CD pipeline was implemented using GitHub Actions and Docker Hub. The pipeline automatically builds the application, executes tests, validates Docker images, and publishes container images. This implementation satisfies the task requirements for automated build and deployment workflows.